

SAT UZBEKISTAN

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1. The expression $\frac{x^{27}(x-7)}{3x^3} + \frac{7x^{27}}{3x^3}$ is equivalent to $\frac{1}{3}x^k$, where k is a constant and $x > 0$. What is the value of k ?
A) 3
B) $\frac{27}{3}$
C) 25
D) 28
2. Megan engages in up to 3 types of exercise each week for a total of 14 hours while training for an ironman competition. Megan runs the same number of minutes each week. The equation $y = 780 - x - 200$ represents the situation where Megan swims for x minutes during a week and bikes for any remaining training time y , in minutes. If this equation is graphed in the xy -plane, which of the following statements is true?
A) During a week when Megan runs for 780 minutes, she bikes for 200 minutes.
B) Each week, Megan swims and bikes for a total of 780 minutes.
C) During a week when Megan doesn't swim, she runs for 260 minutes.
D) During a week when Megan doesn't bike, she swims for 390 minutes.
3. The functions f and g are defined by the equations shown, where a and b are integer constants, $a > b$ and $b > 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or a coefficient, the maximum of the graph of the corresponding function when $x \geq 0$.
I. $f(x) = b(0.97)^{x+a}$
II. $g(x) = b(0.97)^x + a$
A) I only
B) II only
C) I and II
D) Neither I nor II
4. An equilateral triangle is inscribed in a circle with a diameter of 16. Which of the following gives the area of the equilateral triangle?
A) $6\sqrt{3}$
B) $24\sqrt{3}$
C) $32\sqrt{3}$
D) $48\sqrt{3}$
5. The given equation $ax^2 + 88x + c = 0$, where a and c are constants has no real solutions and a factor of $kx + j$. What is the least possible integer value of ac ?
A) 1937 B) 1938 C) 1939 D) 1936
6. $f(t) = 34,000(1.07)^{6t}$
The given function f models the balance of an investment account, in dollars, t years after it is opened. Which statement is the best interpretation of $(1.07)^{6t}$?
A) Every 6 years, the balance increases by 2,380 dollars.
B) Every 6 years, the balance increases by 7% of the previous 6 years balance.
C) Every 2 months, the balance increases by 2,380 dollars.
D) Every 2 months, the balance increases by 7% of the previous 2 months balance.
7. The quadratic function $a(x + 4.5)^2 - d$ can be rewritten as $(x - 9.5)(x + c)$. If a is equal to 1, what is the value of d ?
A) 196 B) 195 C) 194 D) 197

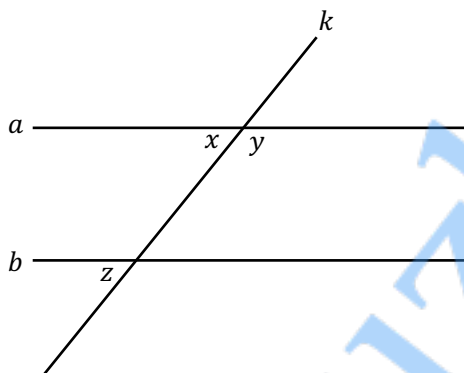
8. $g(x)$ is a quadratic function. In the xy -plane, the graph of $y = g(x)$ passes through the points $(12, 540)$ and $(16, 908)$. If $g(0) = 12$, what is the value of $g(24)$?

A) 1932 B) 1936 C) 1940 D) 1942

9. The surface area of rectangular prism A is 312 m^2 . The surface area of rectangular prism B is $11,232 \text{ m}^2$. Rectangular prism A has a volume of 224 m^3 . If both rectangular prisms are similar, what is the volume of rectangular prism B ?

A) 48384 B) 46384 C) 49384 D) 45384

10.



Note: Figure not drawn to scale.

The given figure shows lines a and b intersected by line k . Which of the following is sufficient to prove that a and b are parallel?

- A) $180 - y = x$
 B) $x = 180 - z$
 C) $y + z = 180 - x$
 D) $180 - z = y$

11.

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(13, 0)$ and $(-6, 0)$. Which of the following is the value of $a + b$ in terms of a ?

- A) $7a$
 B) $6a$
 C) $-6a$
 D) $-7a$

12.

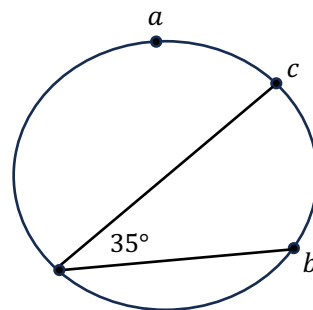
$$f(x) = 1.6^x + 6$$

$$g(x) = 1.8x + b$$

The given system of equations intersects at points (j, k) and (h, r) , where $r < k$. What is the least possible value of b ?

- A) 7
 B) 6
 C) 5
 D) 4

13.



For the given circle, the measure of arc $abc = 130 + 7x$ degrees and the measure of arc $acb = 90 - 3x$ degrees. What is the value of x ?

- A) 52.5 B) 51.5 C) 50.4 D) 60.3

14. The function f is defined by $f(x) = 78(0.39)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. What is the value of p ?

A) 78
B) 61
C) 39
D) 22

15. For which value of a does the quadratic $(7x - 8)^2 + 9 = a$ only have one solution?

A) 9 B) 12 C) 15 D) 18

16. $f(x) = a^x - b$

The exponential function given above passes through points $(c, 12)$ and $(2c, 768)$. What is a possible value of b ?

A) 16 B) 12 C) 8 D) 20

17. If $r = \sqrt[3]{k}$, for the equation $r^{12a-9} = k$, what is the value of a for which r and k are equal?

A) 1 B) 2 C) 3 D) 4

18. What is the perimeter of an equilateral triangle with an area of $289\sqrt{3}$?

A) 102 B) 105 C) 108 D) 111

19. $92x - 1,789 = kx - 1,790$

In the given equation, k is a positive integer constant greater than 91. The equation has exactly one solution. What is the least possible value of k ?

A) 93 B) 92 C) 94 D) 95

- 20.

$$4x^3 + bx^2 - 528x = 0$$

In the given equation, b is a positive integer constant. Which value could be a solution to the equation?

A) -13
B) -12
C) -6
D) 12

- 21.

Circle A has the equation $(x - 18)^2 + y^2 = 27$. Circle B has the equation $(x - 18)^2 + (y - m)^2 = 50$. If Circle B passes through the center of circle A, what is a possible value of m ?

A) $3\sqrt{3}$
B) $5\sqrt{2}$
C) $5\sqrt{2} - 3\sqrt{3}$
D) $5\sqrt{2} + 3\sqrt{3}$

22. If $x^2 = j + k$ and $y^2 = j + m$, which of the following is equal to $(x^2 - y^2)^2$?

A) $k^2 - 2jm + m^2$
B) $k^2 - 2km + m^2$
C) $4j^2 - 4jkm + m^2$
D) $4j^2 - 2jkm + k^2m^2$

23.

$$-x^2 + kx - 3,786$$

In the given equation, k is a positive integer. The equation has no real solution, what is the greatest possible value of k ?

- A) 123 B) 122 C) 121 D) 120

24.

A circle with center $(-7, -4)$ has a radius equal to 14. The equation

$x^2 + y^2 + ax + by + c = 0$ represents the circle. What is the value of c ?

- A) -131 B) -130 C) -132 D) -133

25.

In triangle ABC , angle C is a right angle, point D lies on $\overline{AB}$, point E lies on $\overline{BC}$, and $\overline{DE}$ is parallel to $\overline{AC}$. If the length of $\overline{AB}$ is 39 units, $\overline{DE}$ is 8 units, the length of $\overline{AC}$ is greater than the length of $\overline{BC}$, and the area of triangle ABC is 270 units, what is the length of $\overline{BE}$, in units?

- A) $10/3$ B) 5 C) 4 D) $8/3$

26.

If $k = l + m$, which of the following is equivalent to the expression

$$x^2 - l^2 - 2lm - m^2?$$

- A) $(x + k)^2$
 B) $(x - k)^2$
 C) $(x - k)(x + k)$
 D) $x^2 - kx - k^2$

27.

$$ax^2 + 148x + c$$

In the given expression, a and c are positive constants. If $jx + k$ is a factor of the expression, where j and k are positive constants, what is the greatest possible value of ac ?

- A) 5476 B) 5472 C) 5478 D) 5480

28.

Lines c and d are perpendicular. If the equation of line c is $y = mx + b$ and the lines intersect at point $(7, 6)$. Which point lies on line b ?

- A) $(7 + m, 7)$
 B) $(2m, 8)$
 C) $(7 + 2m, 4)$
 D) $(7 - 3m, 3)$

29.

The given quadratic function

$54x^4 + 219x^2 + 105$ has factors in the form $(k)(ax^2 + b)(cx^2 + d)$. If a, b, c, d , and k are integers, what is the smallest possible value of ab ?

- A) 14 B) 16 C) 18 D) 20

30.

What is the value of k if the given equation only has one solution?

$$\frac{1}{4}|x - 4| - 30 = 2k$$

- A) -15 B) -20 C) -25 D) -30

31. In the given pair of equations a and b are constants. The graph of this pair of equations in the xy -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

$$\begin{aligned} 13x + 17y &= 10 \\ ax + by &= 10 \end{aligned}$$

- A) $39x + 17y = 10$
 $ax - 3by = 10$
- B) $52x + 34y = 10$
 $2ax + 4by = 10$
- C) $65x + 17y = 10$
 $5ax + by = 10$
- D) $13x - 17y = 10$
 $ax + by = 10$

32. Which equation correctly gives c in terms of d , e , and f ?

$$\frac{73}{c} = \frac{73}{d} + \frac{73}{e} + \frac{73}{f}$$

- A) $c = d + e + f$
- B) $c = \frac{def}{d+e+f}$
- C) $c = \frac{def}{d+e+f}$
- D) $c = \frac{def}{de+df+ef}$

33. For the exponential function f , the value of $f(3)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

- A) $f(x) = 3(4)(4)^{x+1}$
- B) $f(x) = 3,072(4)(4)^{x-4}$
- C) $f(x) = 12(4)(4)^x$
- D) $f(x) = 768(4)(4)^{x-3}$

34. A rocket's speed is increasing at a rate of 14.6 meters per second squared. What is the rate in miles per minute squared?
(1 mile = 1,609 meters)

A) 32.67 B) 33.43 C) 32.87 D) 31.84